

```
In [58]: hc_pca = HClust(n_clusters=None,
                        distance_threshold=0,
                        linkage='complete'
                        ).fit(nci_scores[:, :5])
linkage_pca = compute_linkage(hc_pca)
fig, ax = plt.subplots(figsize=(8,8))
dendrogram(linkage_pca,
            labels=np.asarray(nci_labs),
            leaf_font_size=10,
            ax=ax,
            **cargs)
ax.set_title("Hier. Clust. on First Five Score Vectors")
pca_labels = pd.Series(cut_tree(linkage_pca,
                               n_clusters=4).reshape(-1),
                      name='Complete-PCA')
pd.crosstab(nci_labs['label'], pca_labels)
```

12.6 Exercises

Conceptual

- This problem involves the K -means clustering algorithm.
 - Prove (12.18).
 - On the basis of this identity, argue that the K -means clustering algorithm (Algorithm 12.2) decreases the objective (12.17) at each iteration.
- Suppose that we have four observations, for which we compute a dissimilarity matrix, given by



$$\begin{bmatrix} & 0.3 & 0.4 & 0.7 \\ 0.3 & & 0.5 & 0.8 \\ 0.4 & 0.5 & & 0.45 \\ 0.7 & 0.8 & 0.45 & \end{bmatrix}.$$

For instance, the dissimilarity between the first and second observations is 0.3, and the dissimilarity between the second and fourth observations is 0.8.

- On the basis of this dissimilarity matrix, sketch the dendrogram that results from hierarchically clustering these four observations using complete linkage. Be sure to indicate on the plot the height at which each fusion occurs, as well as the observations corresponding to each leaf in the dendrogram.
- Repeat (a), this time using single linkage clustering.
- Suppose that we cut the dendrogram obtained in (a) such that two clusters result. Which observations are in each cluster?
- Suppose that we cut the dendrogram obtained in (b) such that two clusters result. Which observations are in each cluster?

- (e) It is mentioned in this chapter that at each fusion in the dendrogram, the position of the two clusters being fused can be swapped without changing the meaning of the dendrogram. Draw a dendrogram that is equivalent to the dendrogram in (a), for which two or more of the leaves are repositioned, but for which the meaning of the dendrogram is the same.
3. In this problem, you will perform K -means clustering manually, with $K = 2$, on a small example with $n = 6$ observations and $p = 2$ features. The observations are as follows.

Obs.	X_1	X_2
1	1	4
2	1	3
3	0	4
4	5	1
5	6	2
6	4	0

- (a) Plot the observations.
- (b) Randomly assign a cluster label to each observation. You can use the `np.random.choice()` function to do this. Report the cluster labels for each observation.
- (c) Compute the centroid for each cluster.
- (d) Assign each observation to the centroid to which it is closest, in terms of Euclidean distance. Report the cluster labels for each observation.
- (e) Repeat (c) and (d) until the answers obtained stop changing.
- (f) In your plot from (a), color the observations according to the cluster labels obtained.
4. Suppose that for a particular data set, we perform hierarchical clustering using single linkage and using complete linkage. We obtain two dendrograms.
- (a) At a certain point on the single linkage dendrogram, the clusters $\{1, 2, 3\}$ and $\{4, 5\}$ fuse. On the complete linkage dendrogram, the clusters $\{1, 2, 3\}$ and $\{4, 5\}$ also fuse at a certain point. Which fusion will occur higher on the tree, or will they fuse at the same height, or is there not enough information to tell?
- (b) At a certain point on the single linkage dendrogram, the clusters $\{5\}$ and $\{6\}$ fuse. On the complete linkage dendrogram, the clusters $\{5\}$ and $\{6\}$ also fuse at a certain point. Which fusion will occur higher on the tree, or will they fuse at the same height, or is there not enough information to tell?
5. In words, describe the results that you would expect if you performed K -means clustering of the eight shoppers in Figure 12.16, on the basis of their sock and computer purchases, with $K = 2$. Give three answers, one for each of the variable scalings displayed. Explain.